

Department of Computer Science, Gujarat University

MCA-SEM-2

Practical Assignment -1 (Unit-1,2,3,4,7,10)

**Instructions:** 1 to 10 Questions are compulsory for the Assignment submission and 11- 20 questions are Optional (Practical Exam practice Purpose.)

| Serial No. | Questions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| 1          | Develop a <b>menu-driven Java program to manage student records</b> . The program should allow the user to add student details, display student information, and calculate grades based on marks. Implement a <b>Student class with appropriate attributes and constructors</b> , store multiple students using an <b>array of objects</b> , handle invalid marks using a <b>custom exception</b> , and provide options to <b>save and load student data using FileWriter and BufferedReader classes</b> .                                                                                        |
| 2          | Design a <b>menu-driven Java application for an employee payroll system</b> . Create a base class <b>Employee</b> containing common attributes and derive subclasses such as <b>Manager, Developer, and Intern</b> using <b>inheritance</b> . Implement <b>method overriding</b> to calculate salaries differently for each type of employee. The system should allow users to add employees, calculate salaries, and display employee details while handling invalid salary inputs using <b>exception handling</b> , and also allow saving employee data using <b>file handling techniques</b> . |
| 3          | Develop a <b>menu-driven banking system in Java</b> . Create an <b>interface BankOperations</b> with methods such as deposit, withdraw, and checkBalance, and implement it in a class <b>BankAccount</b> . The program should allow users to create accounts, deposit and withdraw money, and display account details. Ensure that <b>invalid withdrawals and insufficient balances are handled using exception handling</b> , and allow storing account details using <b>file handling classes</b> .                                                                                             |
| 4          | Create a <b>menu-driven Java program to calculate areas of different shapes</b> . Implement an <b>abstract class Shape</b> with an abstract method calculateArea(), and derive classes such as <b>Circle, Rectangle, and Triangle</b> that override this method. The program should allow users to select a shape,                                                                                                                                                                                                                                                                                |

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|            | enter required dimensions, and calculate the area using <b>runtime polymorphism</b> , and store the calculated results in a <b>file using file handling classes</b> .                                                                                                                                                                                                                                                                                                                                                        |
| 5          | Design a <b>menu-driven Java application to manage vehicle information</b> . Implement a base class <b>Vehicle</b> with common attributes and derive subclasses <b>TwoWheeler</b> and <b>FourWheeler</b> using <b>inheritance</b> . Demonstrate <b>method overriding</b> to display specific vehicle information. The program should allow users to add and display vehicle details, identify vehicle types dynamically, demonstrate <b>instance and static variables</b> , and allow saving vehicle data to a <b>file</b> . |
| 6          | Develop a <b>menu-driven calculator program in Java</b> that performs operations such as addition, subtraction, multiplication, and division. Create a <b>functional interface Operation</b> containing a method <code>calculate(int a, int b)</code> , and implement the operations using <b>lambda expressions</b> . The program should accept user input, perform operations using a <b>switch statement</b> , and handle division by zero using <b>exception handling</b> .                                              |
| 7          | Create a <b>menu-driven Java program for managing a library system</b> . Implement classes such as <b>Book and Library</b> , where books contain details such as ID, title, and author. The program should allow users to add books, issue books, return books, and display available books. Use <b>arrays of objects</b> to store book records and handle invalid operations using <b>exception handling</b> , and allow saving and loading library data using <b>file handling classes</b> .                               |
| 8          | Design a <b>menu-driven Java application to check voter eligibility</b> . Create a <b>custom exception class named InvalidAgeException</b> that is thrown when the age entered is less than the minimum voting age. The program should allow users to enter voter details, verify eligibility using <b>exception handling</b> , display appropriate messages, and allow storing voter information in a <b>file</b> .                                                                                                         |
| 9          | Develop a <b>menu-driven Java program to maintain an employee database</b> . Implement an <b>Employee class</b> with attributes such as ID, name, and                                                                                                                                                                                                                                                                                                                                                                        |

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|            | <b>salary</b> , and store employee records using <b>RandomAccessFile</b> so that records can be accessed directly. The program should allow users to add employee records, display records, search employees by ID, and update employee information.                                                                                                                                                                                                                                       |
| 10         | Create a <b>menu-driven Java program for a university course registration system</b> . Implement classes such as <b>Student, Course, and Registration</b> , allowing users to add students, add courses, register students for courses, and display registration details. Use <b>arrays of objects</b> , initialize objects using <b>constructors</b> , handle invalid inputs using <b>exception handling</b> , and allow saving registration data using <b>file handling techniques</b> . |
| 11         | Develop a <b>menu-driven Java program for managing sports player statistics</b> . The system should allow users to add player details, display player records, and calculate batting averages. Implement a <b>Player class with constructors</b> , store players using <b>arrays of objects</b> , handle invalid score values using <b>custom exception handling</b> , and allow saving player statistics using <b>file handling classes</b> .                                             |
| 12         | Write a <b>menu-driven Java program to manage a music playlist</b> . Implement a <b>Song class containing attributes such as title, artist, and duration</b> , allow users to add songs, display playlist details, and search songs by title. Handle invalid duration values using <b>exception handling</b> , and allow saving the playlist using <b>file handling classes</b> .                                                                                                          |
| 13         | Design a <b>menu-driven movie ticket booking system in Java</b> that allows users to display available movies, book tickets, cancel tickets, and display booking records. Implement <b>Movie and Ticket classes</b> to manage movie and booking information, validate seat availability using <b>exception handling</b> , and store booking records using <b>file handling techniques</b> .                                                                                                |
| 14         | Develop a <b>menu-driven restaurant food ordering system in Java</b> where users can display menu items, place food orders, and generate the final bill. Implement a <b>FoodItem class containing item name and price</b> , use <b>switch</b>                                                                                                                                                                                                                                              |

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| Serial No. | Questions                                                                                                                                                                                                                                                                                                                                                                                       |
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|            | <b>statements</b> for menu options, and allow storing order details using <b>file handling classes</b> .                                                                                                                                                                                                                                                                                        |
| 15         | Create a <b>menu-driven Java program to manage patient records in a healthcare system</b> . Implement a <b>Patient class containing patient details</b> , allow users to add patients, display patient records, and search patients by ID. Handle invalid patient age values using <b>custom exception handling</b> , and allow saving records using <b>file handling techniques</b> .          |
| 16         | Develop a <b>menu-driven Java application to manage gym membership details</b> . Implement a <b>Member class with constructors</b> , allow users to add member information, display members, and calculate membership fees based on membership duration. Handle invalid duration values using <b>exception handling</b> , and store membership information using <b>file handling classes</b> . |
| 17         | Write a <b>menu-driven Java program for an online course registration system</b> . Implement <b>Student and Course classes</b> , allow users to add courses, register students for courses, and display registration details. Store records using <b>arrays of objects</b> , and allow saving registration data using <b>file handling classes</b> .                                            |
| 18         | Design a <b>menu-driven shopping cart system in Java</b> . Implement a <b>Product class containing product name, quantity, and price</b> , allow users to add products to the cart, display cart items, and calculate the total purchase amount. Handle invalid input values using <b>exception handling</b> , and store cart data using <b>file handling techniques</b> .                      |
| 19         | Develop a <b>menu-driven travel booking system in Java</b> . Implement classes such as <b>Trip and Booking</b> , allowing users to add trip details, book tickets, and display booking information. Initialize objects using <b>constructors</b> , and allow storing booking records using <b>file handling techniques</b> .                                                                    |
| 20         | Create a <b>menu-driven Java fitness tracker system</b> that allows users to add daily activity details, display recorded activities, and calculate calories burned. Implement an <b>Activity class</b> , handle invalid input values using <b>exception handling</b> , and allow storing activity records using <b>file handling classes</b> .                                                 |

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